PHYSICS

Standard Level

Set B | Paper 2

Challenge mock examination

Time allowed: 90 minutes Maximum mark: 50

Candidate name: _____

Candidate number: _____

Date: _____

Instructions

• A calculator and a clean current Physics data booklet are permitted throughout.

• All structured questions are compulsory. There are no optional topics or optional questions.

• Answer all questions. Show physical reasoning, working and units in written responses.

• Use the constants and assumptions given in a question. Otherwise use the data booklet and retain unrounded intermediate values.

• Give numerical answers to an appropriate number of significant figures. Diagrams are schematic unless explicitly stated to be to scale.

• The number in square brackets gives the maximum mark for a written part.

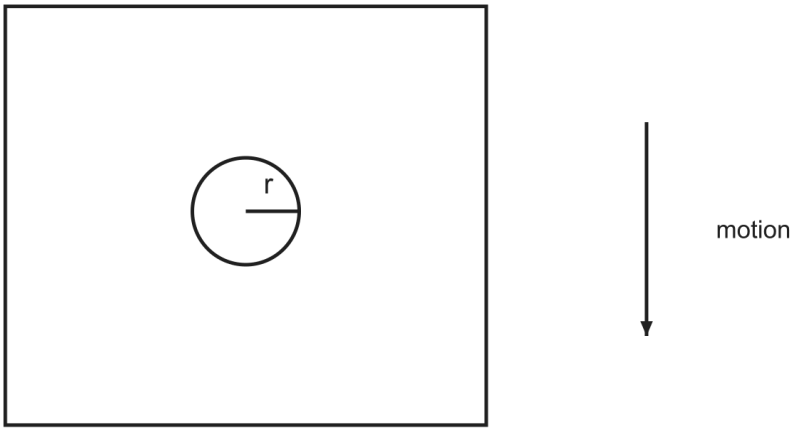
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2 Question 1 | Question paper

Paper 2 - Structured questions

2 Question 1 [8 marks]

A solid sphere of radius $1.00\,\text{mm}$ and density $3000\,\text{kg}\,\text{m}^{-3}$ falls through a stationary liquid of density $1000\,\text{kg}\,\text{m}^{-3}$ and dynamic viscosity $0.200\,\text{Pa}\,\text{s}$. Use $g = 9.81\,\text{m}\,\text{s}^{-2}$. For slow laminar motion, drag magnitude is $6\pi\eta r v$.



liquid vessel; sphere clear of walls

(a) Identify the three forces on the sphere and show that the net downward driving force, excluding drag, is $\frac{4}{3}\pi r^3(\rho_s - \rho_l)g$. [3]

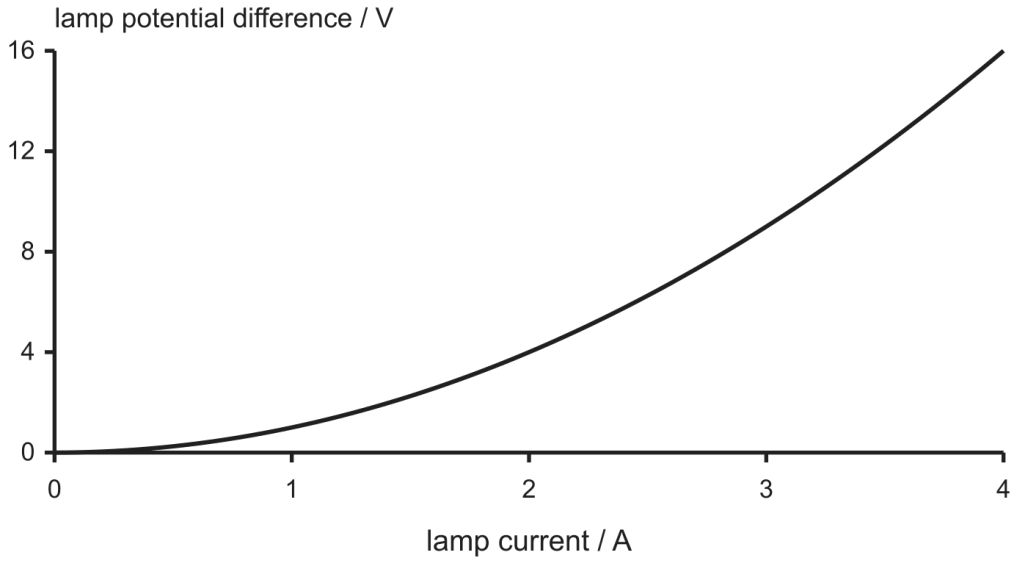
(b) Calculate the terminal speed in this model. [2]

(c) Predict the terminal-speed change if the radius is doubled while all other quantities remain fixed. [2]

(d) State one reason your prediction in (c) might fail for a sufficiently large sphere. [1]

2 Question 2 [8 marks]

A cell of emf $12.0\,\text{V}$ and internal resistance $1.00\,\Omega$ is connected in series with a $2.00\,\Omega$ resistor and a filament lamp. The lamp characteristic is shown and is described by $V_L = I^2$, with V_L in volts and I in amperes. Use the positive operating current.



(a) Explain why the lamp does not obey Ohm's law over this range. [1]

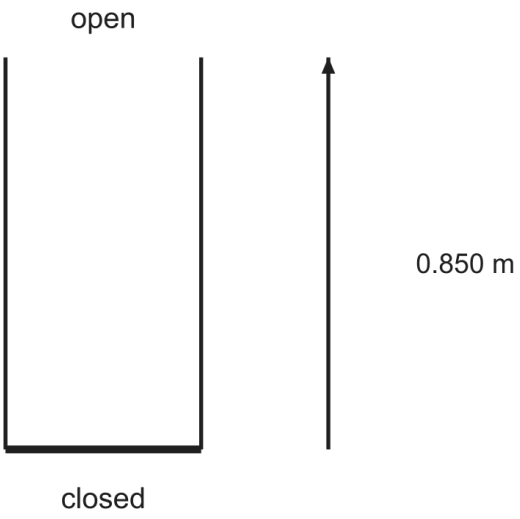
(b) Derive the circuit load-line relation between V_L and I. [2]

(c) Find the operating current, graphically or by calculation. [2]

(d) Calculate the fraction of the cell's electrical power dissipated in the lamp. [3]

2 Question 3 [8 marks]

A uniform tube of physical length $0.850\,\text{m}$ is closed at one end and open at the other. The speed of sound is $343\,\text{m}\,\text{s}^{-1}$. First ignore end correction. The tube is driven by a variable-frequency sound source.



(a) Describe the displacement node and antinode positions in the fundamental mode. [2]

(b) Calculate the fundamental frequency. [2]

(c) Calculate the next allowed resonance frequency. [1]

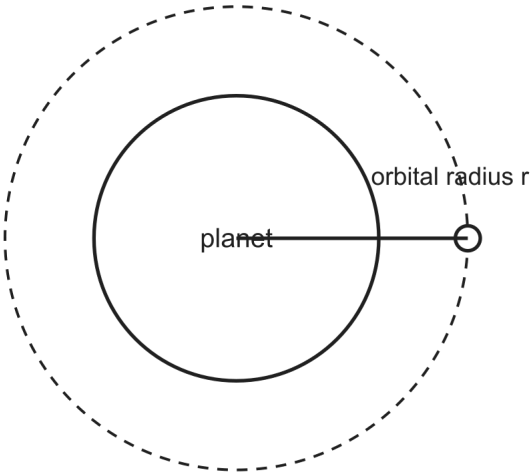
(d) The measured fundamental is $98.6\,\text{Hz}$. Infer the end correction, assuming $L_{\text{effective}} = L_{\text{physical}} + e$. [2]

(e) State the pressure-oscillation condition at the closed end. [1]

2 Question 4 | Question paper

2 Question 4 [8 marks]

A satellite of negligible mass travels in a circular orbit of radius $7.20 \times 10^6\,\text{m}$ about a spherical planet. Its period is $6000\,\text{s}$. The planet radius is $6.00 \times 10^6\,\text{m}$. Use $G = 6.67 \times 10^{-11}\,\text{N}\,\text{m}^2\,\text{kg}^{-2}$. Neglect other bodies.



orbital radius is measured from the centre

(a) Determine the planet mass. [2]

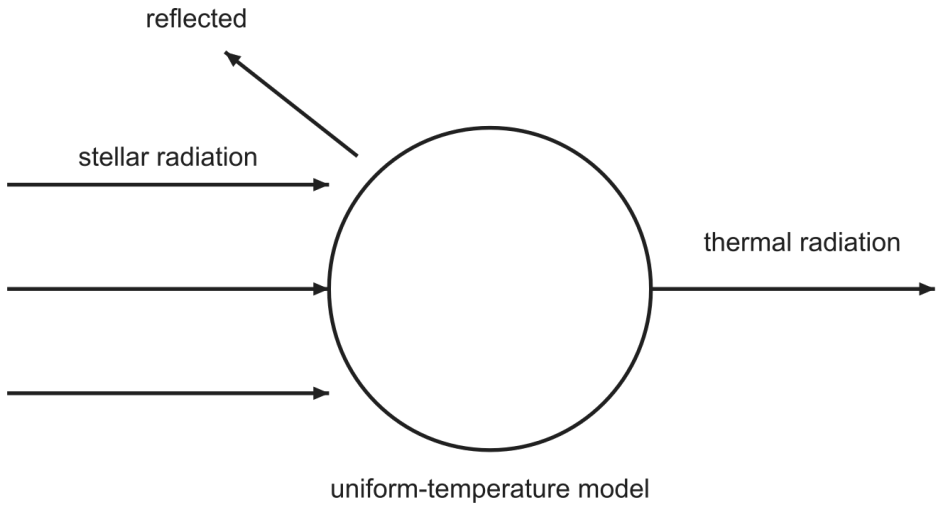
(b) Calculate the satellite orbital speed. [2]

(c) Calculate the gravitational field strength at the planet surface. [2]

(d) A second circular orbit has twice the first orbital radius. Determine its period. [2]

2 Question 5 [9 marks]

A rapidly rotating spherical planet receives stellar intensity $S = 1000\,\mathrm{W\,m^{-2}}$ at its orbit. Its albedo is 0.360 and its effective thermal emissivity is 0.800. Use $\sigma = 5.67 \times 10^{-8}\,\mathrm{W\,m^{-2}\,K^{-4}}$. A simple uniform-temperature model neglects internal heating and has no separate atmospheric layer.



(a) Explain why the globally averaged absorbed intensity is $(1 - \alpha)S/4$. [2]

(b) Calculate the equilibrium temperature. [3]

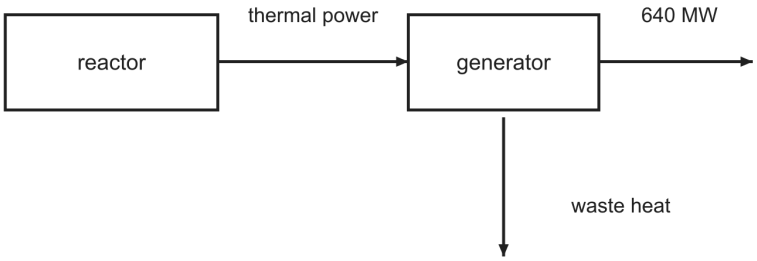
(c) The albedo falls to 0.250 while emissivity remains fixed. Determine the new equilibrium temperature. [2]

(d) Give two limitations of this model when used for a real planet. [2]

2 Question 6 | Question paper

2 Question 6 [9 marks]

A fission power station delivers $640\,\text{MW}$ of electrical power at efficiency 40.0%. Each fission releases $200\,\text{MeV}$. Approximate each fissioned uranium-235 nucleus as having mass $235u$. Use $1\,\text{eV} = 1.60 \times 10^{-19}\,\text{J}$, $u = 1.66 \times 10^{-27}\,\text{kg}$ and one day $= 86400\,\text{s}$.



(a) Explain how the neutron chain reaction can be kept approximately steady. [2]

(b) Calculate the thermal power produced by fission. [1]

(c) Convert the energy released per fission into joules. [1]

(d) Determine the fission rate. [2]

(e) Determine the mass of uranium-235 nuclei undergoing fission per day. [2]

(f) Explain why the mass-equivalent of the released energy is much smaller than the mass calculated in (e). [1]

End of question paper

PHYSICS

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Set B | Paper 2

Worked solution book

Complete the examination before consulting this book. Marking points indicate a method (M), accuracy (A) or reasoning (R) award. The number after the code gives its value. Award only points supported by the candidate's retained working.

Accept valid equivalent approaches and reasonable rounding. Apply error carried forward only where the subsequent method remains physically meaningful; do not reward a contradiction of stated conditions. A correct multiple-choice letter earns one mark; no negative marks apply.

MCQs are automatically marked on the portal. Written marks are self-assessed. Reports show these separately and do not calculate an official IB grade.

Paper 2 - Worked solutions

2 Question 1 [8 marks]

(a) [3 marks]

R1 Weight acts down; buoyancy and motion-opposing viscous drag act up.

M1 Weight is $\frac{4}{3}\pi r^3 \rho_s g$ and buoyancy is the displaced-liquid weight $\frac{4}{3}\pi r^3 \rho_l g$.

A1 Subtracting yields $F_{\text{drive}} = \frac{4}{3}\pi r^3(\rho_s - \rho_l)g$.

(b) [2 marks]

M1 At terminal speed, $6\pi\eta r v_t = F_{\text{drive}}$, so $v_t = 2r^2(\rho_s - \rho_l)g/(9\eta)$.

A1 $v_t = 0.0218\,\text{m}\,\text{s}^{-1}$.

(c) [2 marks]

M1 With the other parameters fixed, the model gives $v_t \propto r^2$.

A1 Doubling radius makes the speed four times as large, $0.0872\,\text{m}\,\text{s}^{-1}$.

(d) [1 mark]

R1 For example, flow may cease to be laminar so Stokes drag no longer applies, or the sphere may become too large for container-wall effects to be negligible. Accept a specific relevant limitation.

2 Question 2 [8 marks]

(a) [1 mark]

R1 $V_L/I = I$ varies with current, so the characteristic does not have constant resistance.

(b) [2 marks]

M1 Apply the loop rule including both the external $2.00\,\Omega$ and internal $1.00\,\Omega$ resistances.

A1 $12.0 = V_L + 3.00I$, so $V_L = 12.0 - 3.00I$.

(c) [2 marks]

M1 Intersect the load line with the lamp curve, or solve $I^2 + 3I - 12 = 0$.

A1 $I = (-3 + \sqrt{57})/2 = 2.27\,\text{A}$. Accept a reasonable graphical value, approximately 2.2 to $2.35\,\text{A}$.

(d) [3 marks]

M1 Lamp power is $P_L = IV_L$; total cell power supplied is εI.

M1 The fraction is $V_L/\varepsilon = I^2/12.0$.

A1 The fraction is 0.431, approximately 43.1%. Use unrounded current, or follow through a reasonable graphical value.

2 Question 3 [8 marks]

(a) [2 marks]

R1 The closed end is a displacement node because the fluid cannot move through it.

R1 The open end is approximately a displacement antinode; the fundamental fits one-quarter wavelength in the effective length.

(b) [2 marks]

M1 $\lambda_1 = 4L$, so $f_1 = v/(4L)$.

A1 $f_1 = 343/(4 \times 0.850) = 101\,\text{Hz}$.

(c) [1 mark]

A1 Only odd harmonics are allowed, so the next resonance is $3f_1 = 303\,\text{Hz}$.

(d) [2 marks]

M1 $L_{\text{effective}} = 343/(4 \times 98.6) = 0.8697\,\text{m}$.

A1 $e = 0.8697 - 0.850 = 0.0197\,\text{m}$, approximately $2.0\,\text{cm}$.

(e) [1 mark]

A1 The closed end is a pressure antinode (maximum pressure-variation amplitude).

2 Question 4 [8 marks]

(a) [2 marks]

M1 $GM = 4\pi^2 r^3/T^2$, so $M = 4\pi^2 r^3/(GT^2)$.

A1 $M = 6.14 \times 10^{24}\,\text{kg}$.

(b) [2 marks]

M1 $v = 2\pi r/T$.

A1 $v = 7.54 \times 10^{3}\,\text{m}\,\text{s}^{-1}$.

(c) [2 marks]

M1 $g_s = GM/R^2$ using the surface radius, not the orbital radius.

A1 $g_s = 11.4\,\text{N}\,\text{kg}^{-1}$.

(d) [2 marks]

M1 For the same planet, $T \propto r^{3/2}$, so $T_2/T_1 = 2^{3/2}$.

A1 $T_2 = 1.70 \times 10^{4}\,\text{s}$.

2 Question 5 [9 marks]

(a) [2 marks]

R1 The planet intercepts sunlight over projected area πR^2 but distributes/radiates it over surface area $4\pi R^2$.

R1 Only fraction $1-\alpha$ is absorbed, giving global average $(1-\alpha)S/4$.

(b) [3 marks]

M1 At equilibrium $(1-\alpha)S/4 = \varepsilon\sigma T^4$.

M1 $T = [0.640(1000)/(4\times 0.800\times 5.67\times 10^{-8})]^{1/4}$.

A1 $T = 244\,\text{K}$ to three significant figures.

(c) [2 marks]

M1 $T_{\text{new}}/T_{\text{old}} = [(1-0.250)/(1-0.360)]^{1/4}$.

A1 $T_{\text{new}} = 254\,\text{K}$ to three significant figures. Retain unrounded values.

(d) [2 marks]

R1 For example, real temperatures differ with latitude, day/night and seasons, so one uniform T need not represent emission exactly.

R1 For example, clouds/atmospheric absorption and emission cannot generally be represented by one fixed emissivity; albedo and emissivity may change with temperature. Award two distinct relevant limitations, not duplicate wording.

2 Question 6 [9 marks]

(a) [2 marks]

R1 On average, each fission must lead to one subsequent fission; too many or too few surviving fission-producing neutrons change the rate.

R1 Neutron-absorbing control rods regulate neutron survival; in a thermal reactor a moderator slows neutrons to increase the likelihood of further fission.

(b) [1 mark]

A1 $P_{\text{thermal}} = 640\,\text{MW}/0.400 = 1.60 \times 10^{9}\,\text{W}$.

(c) [1 mark]

A1 $E_f = 200 \times 10^{6} \times 1.60 \times 10^{-19} = 3.20 \times 10^{-11}\,\text{J}$.

(d) [2 marks]

M1 $\dot{N} = P_{\text{thermal}}/E_f$.

A1 $\dot{N} = 5.00 \times 10^{19}\,\text{s}^{-1}$.

(e) [2 marks]

M1 $m = \dot{N}(86400)(235)(1.66 \times 10^{-27})$.

A1 $m = 1.69\,\text{kg}$ of uranium-235 nuclei per day. This is not the total mass of fuel assemblies.

(f) [1 mark]

R1 Almost all initial mass remains in fission fragments and emitted particles; only the small difference in total rest mass is converted to released energy.

End of worked solution book